Chapter 30 Review

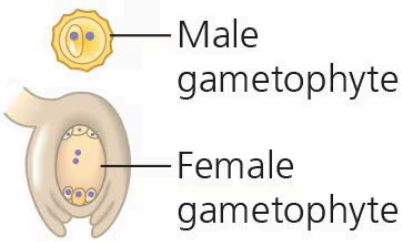
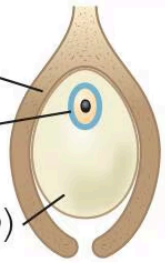
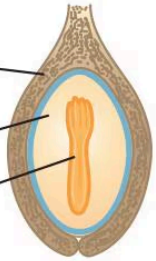
Campbell Biology, AP Edition, 12e

Chapter 30 Review

Summary of Key Concepts

What important changes happened to the mechanisms and structures in the life cycle of angiosperms that gave them greater opportunity to spread geographically?**(BIG IDEAS 2 & 4)**

Concept 30.1: Seeds and pollen grains are key adaptations for life on land

Five Derived Traits of Seed Plants	
Reduced gametophytes	Microscopic male and female gametophytes (*n*) are nourished and protected by the sporophyte (2*n*) 
Heterospory	Microspore (gives rise to a male gametophyte) Megaspore (gives rise to a female gametophyte)
Ovules	Ovule (gymnosperm) { Integument (2*n*), Megaspore (*n*), Megasporangium (2*n*) } 
Pollen	Pollen grains make water unnecessary for fertilization
Seeds	Seeds: survive better than unprotected spores, can be transported long distances { Seed coat, Food supply, Embryo } 

Reduced gametophytes	Microscopic male and female gametophytes (n) are nourished and protected by the sporophyte (2 n)
Heterospory	Microspore gives rise to a male gametophyte. Megaspore gives rise to a female gametophyte.
Ovules	Ovule (gymnosperm) has the following: integument (2 n), megaspore (n), and megasporangium (2 n)
Pollen	Pollen grains make water unnecessary for fertilization
Seeds	Seeds survive better than unprotected spores and can be transported long distances. A seed has a seed coat, food supply, and embryo.

Describe how the parts of an ovule (integument, megaspore, megasporangium) correspond to the parts of a seed.

Concept 30.2: Gymnosperms bear "naked" seeds, typically on cones

- Dominance of the sporophyte generation, the development of seeds from fertilized ovules, and the role of pollen in transferring sperm to ovules are key features of a typical gymnosperm life cycle.
- Gymnosperms appear early in the plant fossil record and dominated many Mesozoic terrestrial ecosystems. Living seed plants can be divided into two monophyletic groups: gymnosperms and angiosperms. Extant gymnosperms include cycads, *Ginkgo biloba*, gnetophytes, and **conifers**.

Although there are just over 1,000 *species of gymnosperms, the group is still very successful in terms of its evolutionary longevity, adaptations, and geographic distribution. Explain.*

Concept 30.3: The reproductive adaptations of angiosperms include flowers and fruits

- ****Flowers** generally consist of four types of modified leaves: **sepals**, **petals**, **stamens** (which produce pollen), and **carpels** (which produce ovules). **Ovaries** ripen into **fruits**, which often carry seeds by wind, water, or animals to new locations.
- Flowering plants originated about 140 million years ago, and by the mid-Cretaceous 100 mya had begun to dominate some terrestrial ecosystems. Fossils and phylogenetic analyses offer insights into the origin of flowers.
- Several groups of **basal angiosperms** have been identified. Other major clades of angiosperms include **magnoliids**, **monocots**, and **eudicots**.

- Pollination and other interactions between angiosperms and animals may have contributed to the success of flowering plants during the last 100 million years.

Explain why Darwin called the origin of angiosperms an "abominable mystery," and describe what has been learned from fossil evidence and phylogenetic analyses.

Concept 30.4: Human welfare depends on seed plants

- Humans depend on seed plants for products such as food, wood, and many medicines.
- Destruction of habitat threatens the extinction of many plant species and the animal species they support.

Explain why destroying the remaining tropical forests might harm humans and lead to a mass extinction.

*For suggested answer, see*Appendix A*.*